Implementing Linear Regression in Python

Python’s Scikit-Learn library provides an easy and efficient way to implement linear regression. Below, we demonstrate how to build and evaluate a linear regression model step by step.

Step 1: Importing Required Libraries

```
from sklearn.model_selection import train_test_split from sklearn.linear_model import
LinearRegression from sklearn.metrics import mean_squared_error
import pandas as pd
```

Step 2: Preparing the Data

Let’s start by creating a dataset that represents the relationship between the square footage of a house and its price: # Example data

```
data = {
    'square_feet': [1500, 2000, 2500, 3000, 3500],
    'price': [300000, 400000, 500000, 600000, 700000]

                                                    }

df = pd.DataFrame(data)
X = df[['square_feet']]
y = df['price']
```

Step 3: Splitting the Data

Splitting the dataset into training and testing subsets ensures that the model can be evaluated on unseen data: # Split data

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Step 4: Training the Model

The LinearRegression class in Scikit-Learn is used to train the model: # Train the model

```
model = LinearRegression()
model.fit(X_train, y_train)
```

Step 5: Making Predictions

Once the model is trained, use it to make predictions on the test set: # Predictions

```
predictions = model.predict(X_test)
```

Step 6: Evaluating the Model